

PART 2

Analytical Line-Fitting Soln.

$$\hat{y} = mx + b$$

$$e_i = y_i - \hat{y}_i$$

$$= y_i - (mx_i + b)$$

$$\text{Points: } (0, 0.5), (2, 3.5), (1, 1.5), (3, 7.5)$$

$$MSE(m, b) = \frac{1}{4} \sum_{i=1}^4 (y_i - (mx_i + b))^2$$

$$= \frac{1}{4} [(0.5 - b)^2 + (3.5 - 2m - b)^2 + (1.5 - m - b)^2 + (7.5 - 3m - b)^2]$$

$$\frac{\partial MSE}{\partial m} = \frac{1}{4} [-4(3.5 - 2m - b) + (-2)(1.5 - m - b) + (-6)(7.5 - 3m - b)]$$

$$= \frac{1}{4} [(-14 + 8m + 4b) + (-3 + 2m + 2b) + (-45 + 18m + 6b)]$$

$$= \frac{1}{4} (28m + 12b - 62)$$

$$\frac{\partial MSE}{\partial m} = 7m + 3b - 15.5 = 0 \text{ @ minimum}$$

$$\frac{\partial MSE}{\partial b} = \frac{1}{4} [(-2)(0.5 - b) + (-2)(3.5 - 2m - b) + (-2)(1.5 - m - b) + (-2)(7.5 - 3m - b)]$$

$$= \frac{1}{4} [(-1 + 2b) + (-7 + 4m + 2b) + (-3 + 2m + 2b) + (-15 + 6m + 2b)]$$

$$= \frac{1}{4} (12m + 8b - 26)$$

$$\frac{\partial MSE}{\partial b} = 3m + 2b - 6.5 = 0 \text{ @ minimum}$$

$$7m + 3b - 15.5 = 0$$

$$3m + 2b - 6.5 = 0$$

$$5m = 11.5$$

$$m = 2.3$$

$$b = -0.2$$

$$\Rightarrow \hat{y} = 2.3x - 0.2$$

x	Actual y	Predicted $\hat{y}$	Error $y - \hat{y}$	Error ²
0	0.5	-0.2	0.7	0.49
2	3.5	4.4	-0.9	0.81
1	1.5	2.1	-0.6	0.36
3	7.5	6.7	0.8	0.64

$$MSE = \sum_{i=1}^4 \frac{\text{Error}_i^2}{4}$$

$$MSE = 0.575$$

Newton-Raphson Soln.

$$\frac{\partial MSE}{\partial m} = 7m + 3b - 15.5$$

$$\frac{\partial MSE}{\partial b} = 3m + 2b - 6.5$$

$$\frac{\partial^2 MSE}{\partial m^2} = 7$$

$$\frac{\partial^2 MSE}{\partial b^2} = 2$$

$$\frac{\partial^2 MSE}{\partial m \partial b} = 3$$

$$\frac{\partial^2 MSE}{\partial b \partial m} = 3$$

$$\nabla MSE = \begin{bmatrix} 7m + 3b - 15.5 \\ 3m + 2b - 6.5 \end{bmatrix}$$

$$H = \begin{bmatrix} 7 & 3 \\ 3 & 2 \end{bmatrix}$$

$$\begin{bmatrix} m_{n+1} \\ b_{n+1} \end{bmatrix} = \begin{bmatrix} m_n \\ b_n \end{bmatrix} - H^{-1} \nabla MSE$$

$$\Rightarrow m_{n+1} = m_n - \frac{\partial MSE / \partial m}{\partial^2 MSE / \partial m^2}$$

$$\Rightarrow b_{n+1} = b_n - \frac{\partial MSE / \partial b}{\partial^2 MSE / \partial b^2}$$

$$\Rightarrow m_{n+1} = m_n - \frac{7m_n + 3b_n - 15.5}{7}$$

$$\Rightarrow b_{n+1} = b_n - \frac{3m_n + 2b_n - 6.5}{2}$$

assume $m_0=0, b_0=0$

$$m_1 = 0 - \frac{7(0) + 3(0) - 15.5}{7}$$

$$m_1 = 2.2$$

$$b_1 = 0 - \frac{3(2.2) + 2(0) - 6.5}{2}$$

$$b_1 = -0.0714$$

$$(m_1, b_1) \approx (2.2, -0.07)$$

$$m_2 = 2.2 - \frac{7(2.2) + 3(-0.07) - 15.5}{7}$$

$$m_2 = 2.2449$$

$$b_2 = -0.0714 - \frac{3(2.24) + 2(-0.0714) - 6.5}{2}$$

$$b_2 = -0.1173$$

$$(m_2, b_2) \approx (2.2449, -0.1173)$$

$$m_3 = 2.2449 - \frac{7(2.2449) + 3(-0.1173) - 15.5}{7}$$

$$m_3 = 2.2646$$

$$b_3 = -0.1173 - \frac{3(2.26) + 2(-0.1173) - 6.5}{2}$$

$$b_3 = -0.1469$$

$$(m_3, b_3) \approx (2.2646, -0.1469)$$

Parabola-Fitting Soln.

$$\hat{y} = ax^2 + b \quad y_i - (ax_i^2 + b)$$

$$MSE(a, b) = \frac{1}{4} \sum_{i=1}^4 [y_i - (ax_i^2 + b)]^2$$

$$MSE(a, b) = \frac{1}{4} [(0.5 - b)^2 + (3.5 - 4a - b)^2 + (1.5 - a - b)^2 + (7.5 - 9a - b)^2]$$

$$\frac{\partial MSE}{\partial a} = \frac{1}{4} [(-8)(3.5 - 4a - b) + (-2)(1.5 - a - b) + (-18)(7.5 - 9a - b)]$$

$$= \frac{1}{4} [(-28 + 32a + 8b) + (-3 + 2a + 2b) + (-135 + 162a + 18b)]$$

$$= \frac{1}{4} (196a + 28b - 166)$$

$$\frac{\partial MSE}{\partial a} = 49a + 7b - 41.5$$

$$\begin{aligned}\frac{\partial \text{MSE}}{\partial b} &= \frac{1}{4} [(-2)(0.5-b) + (-2)(3.5-4a-b) + (-2)(1.5-a-b) + (-2)(7.5-9a-b)] \\ &= \frac{1}{4} [(-1+2b) + (-7+8a+2b) + (-3+2a+2b) + (-15+18a+2b)] \\ &= \frac{1}{4} (28a+8b-26)\end{aligned}$$

$$\frac{\partial \text{MSE}}{\partial b} = 7a+2b-6.5$$

$$49a+7b-41.5=0$$

$$7a+2b-6.5=0$$

$$-7b+4=0$$

$$b=0.5714 \quad a=0.7653$$

$$\hat{y} = 0.7653x^2 + 0.5714$$

x	Actual y	Predicted $\hat{y}$	Error $y-\hat{y}$
0	0.5	0.5714	-0.0714
2	3.5	3.6327	-0.1327
1	1.5	1.3367	0.1633
3	7.5	7.4592	0.0408

$$\text{MSE} = \sum_{i=1}^4 \frac{\text{Error}_i^2}{4}$$

$$\text{MSE} = 0.026$$

$$\frac{\partial \text{MSE}}{\partial a} = 49a+7b-41.5$$

$$\frac{\partial \text{MSE}}{\partial b} = 7a+2b-6.5$$

$$\frac{\partial^2 \text{MSE}}{\partial a^2} = 49$$

$$\frac{\partial^2 \text{MSE}}{\partial b^2} = 2$$

$$a_{n+1} = a_n - \frac{49a_n+7b_n-41.5}{49}$$

$$b_{n+1} = b_n - \frac{7a_{n+1}+2b_n-6.5}{2}$$

assume $a_0=0, b_0=0$

$$a_1 = 0 - \frac{49(0) + 7(0) - 41.5}{49}$$
$$= 0.8469$$

$$b_1 = 0 - \frac{7(0.8469) + 2(0) - 6.5}{2}$$
$$= 0.2857$$

$$a_2 = 0.8469 - \frac{49(0.8469) + 7(0.2857) - 41.5}{49}$$
$$= 0.806$$

$$b_2 = 0.285 - \frac{7(0.806) + 2(0.2857) - 6.5}{2}$$
$$= 0.4286$$

$$a_3 = 0.806 - \frac{49(0.806) + 7(0.143) - 41.5}{49}$$
$$= 0.7857$$

$$b_3 = 0.143 - \frac{7(0.7857) + 2(0.143) - 6.5}{2}$$
$$= 0.5$$

$$(a, b) = (0.7857, 0.5)$$